

GATE Roadmap & Opportunities Guide for a B.Tech CSE Student

A Comprehensive, Fact-Checked Career Framework

Target Audience: 2nd-Year B.Tech Computer Science Engineering
Data Context: GATE 2025/2026 Cycle | Last Updated: September 2026

Introduction & Disclaimer

This report is engineered specifically for a 2nd-year B.Tech Computer Science and Engineering (CSE) student. It maps out exactly what the Graduate Aptitude Test in Engineering (GATE) is, the pathways it unlocks, and how to optimally prepare over a 3-year timeline.

Important Note on Year-Dependent Data

GATE rules, combinations, Public Sector Undertaking (PSU) vacancies, and cutoffs are subject to change annually. The information provided here is based on the

current official GATE 2025/2026 organizing rules (e.g., IIT Roorkee for 2025)

, COAP, and CCMT guidelines. Always verify with official brochures in the year you appear for the exam.

Section 1: What is GATE?

The **Graduate Aptitude Test in Engineering (GATE)** is a national-level examination that tests the comprehensive understanding of various undergraduate subjects in engineering and science. For 2025, it is organized by **IIT Roorkee** on behalf of the National Coordination Board (NCB).

- **Why Take GATE?** It is the primary gateway for Master's programs (M.Tech/MS/PhD) at elite Indian institutes (IITs, IISc, NITs) with financial assistance, and for direct executive recruitment in leading PSUs.
- **Score Validity:** A GATE score is valid for **THREE YEARS** from the date of announcement of results for M.Tech admissions. However, for PSU recruitment, the score is typically valid for **ONLY ONE YEAR** (the specific recruitment year).
- **Age Limit:** There is **NO age limit** to appear for GATE.
- **Eligibility Rules (Crucial for 2nd Year):** Candidates who are currently studying in the **3rd year or higher** of any undergraduate degree program, or have already completed any government-approved degree in Engineering/Technology, are eligible.

Can a 2nd-year student appear?

No. As a 2nd-year B.Tech student, you **cannot**

register for or appear in GATE right now. You will be eligible to appear for the first time in February of your

3rd year (6th Semester)

. The current time (2nd year) is meant purely for building your fundamental concepts.

Section 2: All GATE Papers

GATE is conducted in 30 different subject papers (as of the latest syllabus). Candidates can choose to appear in one or two papers. For a CSE student, the most relevant papers are **CS (Computer Science and Information Technology)** and **DA (Data Science & Artificial Intelligence)**.

Code	Paper Name	Relevance for CSE Student
CS	Computer Science and Information Technology	Core relevance. Essential for all standard CSE M.Tech and PSU jobs.
DA	Data Science and Artificial Intelligence	High relevance. Excellent secondary paper for AI/ML specific M.Tech programs.
EC	Electronics and Communication Engineering	Low relevance. Only if pivoting to hardware/VLSI.
MA	Mathematics	Moderate. Good for heavily theoretical research.
ST	Statistics	Moderate. Good for data-centric theoretical research.
GE	Geomatics Engineering	Low. Related to GIS/spatial data.
PH	Physics	Low relevance.

Other Papers (Low/No Relevance to CSE): AE (Aerospace), AG (Agricultural), AR (Architecture), BM (Biomedical), BT (Biotechnology), CE (Civil), CH (Chemical), CY (Chemistry), EE (Electrical), ES (Environmental), EY (Ecology), GG (Geology), IN (Instrumentation), ME (Mechanical), MN (Mining), MT (Metallurgical), NM (Naval), PE (Petroleum), PI (Production), TF (Textile), XE (Engineering Sciences), XH (Humanities), XL (Life Sciences).

Section 3: GATE CS for CSE Students

CS (Computer Science & Information Technology) is the natural choice for a B.Tech CSE student because its syllabus aligns almost perfectly with the standard 2nd and 3rd-year university curriculum.

Subject-Wise Historical Weightage & Strategy

(Note: Official weightages are not strictly fixed except for Aptitude, but historical trends provide reliable estimates out of 100 marks).

Subject	Approx. Weightage	Strategic Difficulty (General Consensus)
General Aptitude	15 Marks	Scoring. High ROI for practice.
Engineering & Discrete Math	13-15 Marks	Moderate. Requires heavy practice. Discrete is foundational.
Programming & Data Structures	10-12 Marks	Scoring. Highly logical, easy if you code regularly.
Algorithms	6-8 Marks	Moderate to Hard. Tricky theoretical complexity questions.
Theory of Computation (TOC)	8-10 Marks	Scoring. Very logical, finite patterns. Highly rewarding.
Compiler Design	4-6 Marks	Scoring. Short syllabus, predictable questions.
Digital Logic	4-6 Marks	Scoring. Formulaic and deterministic.
Computer Organization & Arch (COA)	8-10 Marks	Hard. Vast, hardware-oriented, confusing wording in PYQs.

Operating Systems (OS)	8-10 Marks	Moderate. Conceptual, requires deep understanding of synchronization.
Databases (DBMS)	6-8 Marks	Scoring. SQL, Normalization, and Transactions are straightforward.
Computer Networks (CN)	8-10 Marks	Hard. Extremely vast syllabus, requires memorization and numerical practice.

Strategic Prerequisite Order

- **Math before everything:** Discrete Mathematics contains Logic and Set Theory, essential for TOC, DBMS, and Algorithms.
- **The Code Pipeline:** C Programming → Data Structures → Algorithms.
- **The Hardware Pipeline:** Digital Logic → Computer Organization (COA) → Operating Systems.
- **The Theory Pipeline:** Theory of Computation (TOC) → Compiler Design.

Section 4: The Two-Paper Option

The GATE authorities allow candidates to appear in up to **two papers** to widen their postgraduate opportunities. The combinations are pre-determined by the organizing institute every year.

Current Allowed Combinations for CS (Based on 2025 Rules)

If your primary paper is **CS**, your allowed secondary options are: **DA, EC, GE, MA, PH, ST**.

1. CS + DA (Data Science & AI) → HIGHLY RECOMMENDED

The DA paper was introduced recently. The syllabus shares ~50% overlap with CS (Programming, DS, Algo, Linear Algebra, Calculus, Probability). You only need to study Machine Learning and AI additionally. Taking DA opens doors exclusively for specialized M.Tech in AI/Data Science at top IITs.

2. CS + MA (Mathematics) or ST (Statistics)

Only recommended if you wish to pursue hardcore theoretical research in computational mathematics or statistics. Negligible benefit for standard software placement-focused M.Techs.

3. CS + EC / GE / PH

Generally unnecessary for a standard CSE student. Syllabus overlap is extremely low, leading to split focus and poor scores in both.

Does a second paper automatically mean more M.Tech options?

No.

Your eligibility for an M.Tech program is dictated by the institute's specific rules. An IIT might say "M.Tech in CSE requires B.Tech in CSE + GATE in CS". In this case, a DA score wouldn't help. However, for "M.Tech in AI", an IIT might accept *both* CS and DA scores. Writing a second paper is

unnecessary

if your sole aim is a standard CSE M.Tech or a core PSU job (which strictly demands CS).

Section 5: What "GATEs" Open After GATE? (Visual Overview)

A GATE CS score acts as a master key for multiple distinct career pathways. Below is the comprehensive map.

Pathway A & B: IITs and IISc (The Premier Tier)

What GATE does: Shortlists you.

What it does NOT guarantee: Admission. Many top programs have a subsequent written test and/or interview.

Process: Apply individually to each institute → Receive offers via COAP portal.

Pathway C, D, E, F: NITs, IIITs, IISERs, IIST

What GATE does: Directly dictates admission (mostly direct seat allocation based on score).

Process: Centralized counselling through CCMT portal.

Pathway G & H: CFTIs, State/Private/Deemed Universities

Examples: HCU, JNU, DTU, NSUT, BITS (some campuses accept GATE for stipends), Jadavpur University.

Process: Separate university-specific applications and counselling.

Pathway I & J: M.Tech / MS (Research) / Direct PhD

M.Tech: 2-year course-heavy program. High placement focus.

MS (Research): 2.5 to 3-year research-heavy program. Lower cutoff than M.Tech, requires an interview.

Direct PhD: 5-year integrated program directly after B.Tech for highly motivated researchers.

Pathway K: Scholarships & Stipends

GATE qualification guarantees an AICTE/MoE stipend (currently ₹12,400/month) if admitted to an approved master's program.

Pathway L: PSU Recruitment

Direct executive-level jobs in government enterprises (e.g., ONGC, IOCL, DRDO). Separate applications based on GATE scores, followed by interviews.

Pathway M: Foreign Universities

Admissions to postgraduate programs in Singapore (NUS, NTU) and Germany (TU Munich, RWTH Aachen).

Section 6: IIT / IISc Admissions via COAP

Admissions into premier institutes are decentralized but coordinated. You must understand the distinction between applying and accepting offers.

What is COAP?

COAP (Common Offer Acceptance Portal) is a common platform where all participating IITs and IISc upload their admission offers.

What COAP is NOT: It is **NOT** an application portal. You cannot "apply for IIT Bombay" on COAP. You must visit the IIT Bombay website, pay the fee, submit your GATE score, and apply. Later, if IIT Bombay selects you, they will push the offer to your COAP dashboard, where you can "Accept & Freeze", "Retain & Wait", or "Reject".

The Admission Funnel

1. **GATE Score:** Your normalized score out of 1000.
2. **Cutoff / Shortlist:** The institute decides a minimum score (e.g., 750) to call students for the next stage.
3. **Written Test / Interview:** Institutes like IISc (for AI/CSE) or IIT Bombay (for TA/RA categories) often conduct heavy coding tests and technical interviews.
4. **Final Admission:** Based on a composite score (e.g., 70% GATE + 30% Interview).

Why "X rank guarantees IIT Y" is false:

Because of the interview stage for MS/Research programs and specialization-specific criteria, a student with AIR 150 might get rejected after a poor interview, while a student with AIR 400 might secure the seat by acing the test. Only direct M.Tech (Teaching Assistantship - TA) programs rely mostly on direct GATE scores.

Section 7: NIT / IIIT / CFTI Admissions (CCMT)

Unlike IITs, the admission process for NITs, IIITs, and Centrally Funded Technical Institutes (CFTIs) is highly centralized through a portal called **CCMT (Centralized Counselling for M.Tech/M.Arch/M.Plan)**.

- **How it works:** You register on CCMT once, pay a single counselling fee, and create a preference list of institutes and programs (e.g., 1. NIT Trichy CSE, 2. NIT Surathkal CSE, 3. IIIT Allahabad IT...).
- **Seat Allotment:** An algorithm allocates seats strictly based on your GATE score, category, and preference list. There are usually no interviews for standard M.Tech programs via CCMT.
- **Upgradation & Reporting:** Across 3-4 rounds, you can float (upgrade) your seat. Once finalized, you must verify documents online, pay the seat acceptance fee, and physically report to the allotted institute.

Note: Participating institutes change slightly each year. Always refer to the latest CCMT brochure.

Section 8: PSU Opportunities for CSE Students

Public Sector Undertakings (PSUs) offer highly lucrative, stable government jobs. However, CSE students have **fewer PSU vacancies compared to Electrical or Mechanical branches**. It is vital to maintain realistic expectations.

PSU / Organization	Sector	Recruits CS via GATE?	Typical Profile / Next Stage
ONGC	Energy	Yes Frequent	Programming Officer. Shortlist via GATE → Interview.
IOCL	Energy	Yes Frequent	Information Systems Officer. Shortlist via GATE → GD + Interview.
DRDO (RAC)	Defence R&D	Yes Frequent	Scientist 'B'. GATE Score triggers Descriptive Exam → Interview.
BARC (OCES)	Nuclear R&D	Yes Frequent	Scientific Officer. GATE Score OR separate BARC exam → Heavy Technical Interview.
POSOCO (Grid-India)	Energy Grid	Yes Frequent	Executive Trainee (CS). GATE → GD + Interview.

AAI	Aviation	Yes Sometimes	Junior Executive (IT). Directly via GATE score / Interview.
NTPC / POWERGRID	Energy	Rare	Historically focus on EE/ME/CE. CS recruitment via GATE is infrequent and highly variable.
BSNL / ECIL	Telecom/ Elec	Varies	Often conduct their own separate exams, but occasionally use GATE.

Crucial PSU Reality Check:

Never assume a PSU will definitely recruit CSE in the year you graduate. Wait for their official recruitment notification. Even when they do, the number of UR (Unreserved) vacancies for CSE is usually single or double digits across the country. You typically need an

AIR under 150

to secure a PSU interview call.

Section 9: GATE + Scholarships & Stipends

A massive advantage of qualifying GATE is financial independence during your postgraduate studies.

- **Current Stipend Amount:** ₹12,400 per month (Standardized by AICTE / MoE).
- **Duration:** 24 months for M.Tech/ME programs.
- **Eligibility:** You must have a **valid GATE score** (i.e., above the qualifying cutoff for your category) AND be admitted to an AICTE/MoE-approved institution.
- **Conditions:** Students receiving this stipend are classified as Teaching Assistants (TA). You are required to assist the department for **8 hours per week** (grading papers, lab duties, invigilation).
- **Difference between Qualification and Stipend:** Simply "qualifying" GATE does not deposit money into your account. You must secure admission into an M.Tech program. The institute then applies for the stipend on your behalf.

Section 10: Foreign Universities Accepting GATE

GATE is internationally recognized by a select group of elite technical universities, primarily in Singapore and Germany. This provides an alternative to the GRE for Indian students.

- **Singapore:**

- **National University of Singapore (NUS) & Nanyang Technological University (NTU)**
- **Programs:** M.Eng, M.Sc, and PhD programs.
- **Requirement:** Extremely competitive. Usually demands a GATE percentile of **90% or above** (roughly AIR < 10,000, though realistically top 1,500 ranks have better chances), an excellent B.Tech CGPA, and TOEFL/IELTS scores.

- **Germany:**

- **TU Munich (TUM) & RWTH Aachen**
- **Programs:** Various English-taught M.Sc Engineering programs.
- **Requirement:** GATE serves to fulfill subject-competency requirements for Indian bachelor degree holders. High GATE scores improve admission chances significantly. IELTS/TOEFL is mandatory.

Do not overstate this pathway:

GATE is an *acceptable* substitute for GRE in these specific institutes, but it does not bypass their rigorous academic GPA or language proficiency requirements.

Section 11: Rank vs Score vs Marks (Demystified)

GATE results often confuse beginners. Here is how to read them:

- **Raw Marks:** The actual marks you calculate out of 100 based on the answer key (e.g., 62.33 / 100).
- **Normalized Marks:** Because GATE CS usually has huge applicant numbers, it is conducted in multiple shifts (Session 1 and Session 2). A mathematical formula normalizes marks to ensure fairness if one paper was harder.
- **GATE Score:** A standardized score calculated out of 1000. It maps your performance relative to the topper and the qualifying marks. Institute cutoffs (IITs/NITs) are *always* published as GATE Scores (e.g., 750/1000), not marks.
- **AIR (All India Rank):** Your absolute position nationally among all CS test takers.

- **Qualifying Cutoff:** The minimum marks (out of 100) needed to "pass" GATE. Usually ~25-29 marks for General category in CS. Getting this only means you get a scorecard; it guarantees zero admissions.

Example: In 2024, 60 marks might yield an AIR of 1000. In 2025, if the paper is much harder, 60 marks might yield an AIR of 300. This is why *GATE Score* out of 1000 remains a stable metric for admissions across years, whereas Marks fluctuate wildly.

Section 12: Practical CSE Target Checklist

Based on historical trends (General Category), here is a realistic mapping of Ranks to Opportunities. *(Note: OBC/EWS/SC/ST categories have significantly more relaxed cutoffs, often extending these rank brackets by 2x to 5x depending on the tier).*

Target Goal	Approx. AIR Target (General)	Likely Outcomes
Just Qualify	Top 15,000 - 20,000	GATE Scorecard. Eligible for private/state colleges and basic stipends. No premier institutes.
Moderate / Tier-2 NITs	AIR 1,500 - 3,500	Newer NITs, lower-tier IITs, centralized CFTIs.
Strong NITs / New IITs	AIR 500 - 1,500	Top 3 NITs (Trichy, Surathkal, Warangal), New IITs (Hyderabad, Gandhinagar, Indore), MS Research at older IITs.
Top IITs / IISc M.Tech	AIR 100 - 500	Direct M.Tech at Top 7 IITs (Bombay, Delhi, Madras, Kanpur, Kharagpur, Roorkee, Guwahati) and IISc.
PSUs / Elite Elite Guarantee	AIR 1 - 100	Direct interview calls for ONGC, IOCL, POSOCO. Utter dominance in IIT choices.

Section 13: Complete GATE CS Syllabus Checklist

Use this as your ultimate tracking sheet over the next two years. (*Base authority: Official GATE Syllabus*).

- **1. Engineering Mathematics:** Discrete Mathematics (Propositional logic, sets, relations, functions, graphs, combinatorics), Linear Algebra (Matrices, determinants, eigenvalues/vectors), Calculus (Maxima/minima), Probability (Conditional, mean, median, variance, distributions).
- **2. Digital Logic:** Boolean algebra, combinational and sequential circuits, minimization, number representations and computer arithmetic (fixed/floating point).
- **3. Computer Organization & Architecture:** Machine instructions, addressing modes, ALU, data path and control unit, instruction pipelining, pipeline hazards, memory hierarchy (cache, main memory), I/O interface.
- **4. Programming and Data Structures:** Programming in C, recursion, arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs.
- **5. Algorithms:** Searching, sorting, hashing, asymptotic worst-case time/space complexity, algorithm design techniques (greedy, dynamic programming, divide-and-conquer), graph traversals, shortest paths, MST.
- **6. Theory of Computation:** Regular expressions, finite automata, context-free grammars, push-down automata, regular and context-free languages, pumping lemma, Turing machines, undecidability.
- **7. Compiler Design:** Lexical analysis, parsing, syntax-directed translation, runtime environments, intermediate code generation, local optimization.
- **8. Operating Systems:** System calls, processes, threads, inter-process communication, concurrency, synchronization, deadlock, CPU and I/O scheduling, memory management, virtual memory.
- **9. Databases:** ER-model, relational model, relational algebra, tuple calculus, SQL, integrity constraints, normal forms, file organization, indexing (B and B+ trees), transactions, concurrency control.
- **10. Computer Networks:** OSI/TCP-IP stacks, LAN (Ethernet, Token Ring), ALOHA, switching, routing algorithms, IPv4/IPv6, TCP/UDP, congestion control, Application layer protocols (DNS, SMTP, HTTP, FTP).
- **11. General Aptitude:** Verbal aptitude, quantitative aptitude, analytical aptitude, spatial aptitude.

Section 14 & 15: The 2nd-Year to GATE Roadmap & Study Plan

Phase 1: Year 2 (Semester 3 & 4) - The Foundation

Goal: Master the subjects currently being taught in your B.Tech curriculum. Do not rush to 4th-year subjects.

Subjects to Prioritize: C Programming, Data Structures, Algorithms, Digital Logic, Discrete Mathematics.

What NOT to worry about: Mock test series, admission forms, complex OS/CN concepts (unless taught in college).

Daily Plan (1-2 Hours/Day): Pay extreme attention to college lectures for core subjects. Dedicate 1 hour in the evening to solve basic GATE Previous Year Questions (PYQs) corresponding strictly to the topic taught in college that week.

Phase 2: Year 3 (Semester 5 & 6) - The Core Grind

Goal: Complete 80% of the syllabus.

Subjects: OS, DBMS, TOC, Compiler Design, COA, CN.

Summer (Post-2nd Year): Consolidate Programming, DS, Algo, Math.

Action: Register and appear for GATE in Feb of your 3rd year for the experience. Buy a standard topic-wise mock test series.

Phase 3: Year 4 (Semester 7 & 8) - Optimization & Execution

Goal: Rank refinement.

Action: Identify weak areas from your 3rd-year GATE attempt. Finish remaining subjects. Shift entirely to Full-Length Mock Tests by November. Manage B.Tech placements simultaneously (GATE prep naturally helps with core CS placements).

Consistency Matrix: Weekly Study Plans

- **1-2 Hours/Day (Ideal for 2nd Year):**

1 hr learning concept + 30 mins standard textbook problems + 30 mins solving 5-10 GATE PYQs on that specific concept.

- **3-4 Hours/Day (Ideal for 3rd/4th Year):**

2 hrs video lectures/notes + 1 hr rigorous PYQ solving + 1 hr revision of older subjects/short notes.

Do not encourage unhealthy overworking. A steady 2 hours a day from 2nd year will crush a panic-driven 10 hours a day in 4th year.

Section 16: Optimal Subject Order

Never study randomly. Follow the dependency chains:

1. **Math Foundation:** Discrete Mathematics (do this first, it builds pure logical thinking).
2. **Software Track:** C Programming → Data Structures → Algorithms.
3. **Hardware Track:** Digital Logic → Computer Organization (COA) → Operating Systems.
4. **Systems Track:** TOC → Compiler Design.
5. **Independent Tracks:** DBMS, Computer Networks, Aptitude.

Section 17: PYQ Strategy (The Holy Grail)

- **When to start:** Immediately after finishing a single topic. Do NOT wait until the whole subject is over.
- **How many years:** Last 15-20 years for CS.
- **The Error Notebook:** Maintain a dedicated diary. Every time you get a PYQ wrong, write down the *concept* you missed, not just the solution. Revisit this notebook every Sunday.

Section 18: Mock Test Strategy

- **Topic Tests:** Take these in 3rd year as soon as a chapter is done.
- **Subject Tests:** Take these to stitch chapters together.
- **Full-Length Mocks:** Start strictly by December of your final attempt year. Attempt them in the exact time slot of the real GATE exam (e.g., 9:30 AM - 12:30 PM).
- **Analysis:** Spending 3 hours giving a mock test demands 2 hours of post-test analysis. Look for accuracy errors vs. conceptual errors.

Section 19: Deadly Mistakes to Avoid

- **Ignoring Aptitude and Math:** Together they form ~28 marks. You cannot get a top rank by ignoring a quarter of the paper.
- **Memorizing without understanding:** GATE rarely repeats exact questions; it repeats patterns. Formula memorization fails in CS.
- **Starting mocks too late:** Fear of "incomplete syllabus" delays mocks. Give mocks even with 70% syllabus done.
- **Chasing random blogs/rank predictors:** Stick to official cutoffs. Cutoffs fluctuate based on paper difficulty.
- **Ignoring College Academics:** A bad CGPA (below 7.0 or 7.5) will disqualify you from IIT/PSU interviews regardless of your GATE rank. Keep college scores high!

Section 20: Career Decision Tree (After GATE)

Branch 1: GATE Score is Extremely Strong (AIR 1 - 300)

- **Top Choice:** Direct M.Tech in CSE/AI at Top 5 IITs or IISc (Incredible placements, 20LPA+ avg).
- **Alternative:** PSU Executive Recruitment (ONGC, IOCL) if vacancies exist and you desire stability + high perks.
- **Alternative:** Direct PhD at IISc for hardcore R&D.

Branch 2: GATE Score is Good (AIR 300 - 1500)

- **Top Choice:** M.Tech at New IITs (Hyderabad, Gandhinagar) or Top NITs (Trichy, Surathkal, Warangal). Placements are still exceptional.
- **Alternative:** MS by Research at Top IITs (requires cracking an interview).

Branch 3: GATE Score is Moderate (AIR 1500 - 4000)

- **Top Choice:** M.Tech at Mid-tier NITs or IIITs (Allahabad, Gwalior) via CCMT.
- **Alternative:** Drop a year for a better rank if highly ambitious, or accept college campus placements and prepare while working.

Branch 4: Did not qualify / Poor Score

- B.Tech Placements (Software Engineering).
- Higher Studies Abroad (via GRE/IELTS/TOEFL).
- Pivot to MBA (CAT) or Civil Services (UPSC).

Section 21: Official Verification & Links

Always trust official sources over coaching institutes or random blogs. Click the links below for primary data:

- [Official GATE 2025 Website \(IIT Roorkee\)](#)
- [CCMT \(NIT/IIIT Admissions Portal\)](#)
- [COAP \(IIT Offer Portal - Link updates annually\)](#)
- [IISc Bangalore Admissions](#)
- [ONGC Official Careers](#)

Section 22: One-Page Summary Cheat Sheet

GATE 2025/2026 CS Quick Facts

Primary Paper: Computer Science & Information Technology (CS)

Best Secondary Paper: Data Science & AI (DA)

Eligibility: 3rd Year B.Tech and above. (You will appear in Feb of 3rd year).

Validity: 3 Years for Academics, 1 Year for PSUs.

Stipend: ₹12,400/month via AICTE (Requires admission, not just clearing cutoffs).

Pattern: 65 Questions, 100 Marks, 3 Hours. (MCQ, MSQ, NAT types). Negative marking only for MCQs.

Portals: Apply individually to IITs → accept on **COAP**. Apply centrally to NITs → allocate via **CCMT**.

PSU Reality: CSE vacancies are limited but exist in ONGC, IOCL, DRDO, BARC. Requires top 100 rank.

2nd Year Action Item: Do NOT panic. Focus strictly on Discrete Math, C, Data Structures, and Algorithms. Build an impregnable foundation now, and solve topic-wise PYQs daily.